

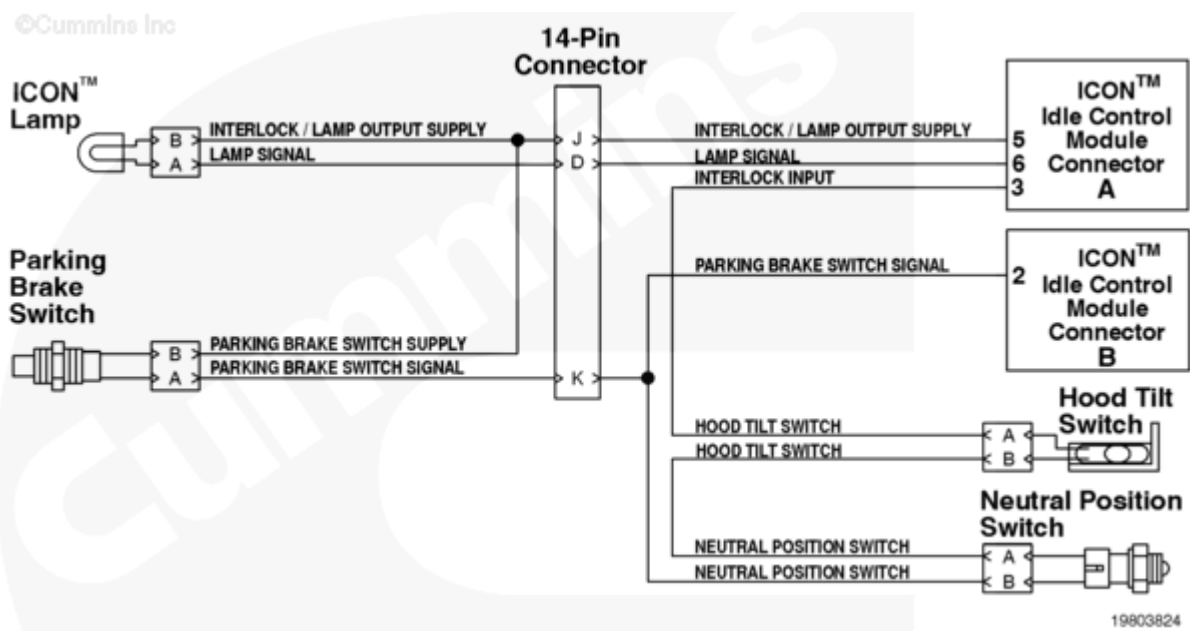
Fault Code: 767 (Aftermarket and OEM)

Starter Interlock Safety Circuit - Voltage Below Normal or Shorted to Low Source

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 767 PID(P): SPN: FMI: Lamp: SRT:	Starter Interlock Safety Circuit - Voltage Below Normal or Shorted to Low Source. Less than 6-VDC pulse detected at the interlock lamp output circuits when high voltage was expected by the ICON™ idle control module.	The ICON™ system will be disabled. Only mandatory shutdown will be enabled. Engine will start normally.



Circuit Description

The interlock input and lamp output circuit provides power to the ICON™ lamp and to the interlock switches. The above circuit diagram can vary, such as connector or pins, depending on the vehicle make or model. OEM installations can possibly provide the harnessing between the idle control module and other ICON™ devices.

Component Location

The ICON™ lamp is typically located in the vehicle cab on the dash panel. The parking brake switch is typically located behind the dash on the parking brake air line. The hood tilt switch is typically located on the hood behind the left headlight housing. The neutral position switch is located on the top cover plate of the transmission near the gear shift.

The ICON™ module can be located in a different location depending on the vehicle application.

Shoptalk

This fault indicates a short circuit to ground. This circuit outputs a pulse width modulated (PWM) voltage at pin 5 of the ICON™ idle control module A connector.

The ICON™ system can display **only** the present active fault code. If more than one fault code is active at the same time, the ICON™ system flashes out the highest priority fault. After the fault has been corrected then the next active fault will be flashed out.

Note: The ICON™ electronic service tool can display more than one active and or inactive fault codes at the same time.

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new ICON™ idle control module, all other active fault codes must be investigated prior to replacing the ICON™ idle control module.

CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:

Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead

Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Read the fault codes.	

STEPS		SPECIFICATIONS
	STEP 1A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes.	Fault Code 767 active
STEP 2.	Check the ICON™ idle control module.	
	STEP 2A. Check the ICON™ idle control module A connector for damaged pins.	No damaged pins
	STEP 2B. Check the hood tilt switch for a short circuit to ground.	More than 100k ohms
	STEP 2C. Check the ICON™ components for a short circuit to ground.	More than 100k ohms
STEP 3.	Identify the ICON™ system.	
	STEP 3A. Identify whether the ICON™ system is an Aftermarket or an OEM.	ICON™ system is Aftermarket
STEP 4.	Check the ICON™ engine harness.	
	STEP 4A. Inspect the ICON™ engine harness 14-pin connector pins.	No damaged pins
	STEP 4B. Check the ICON™ engine harness for a short to ground.	More than 100k ohms
STEP 5.	Check the ICON™ cab harness.	
	STEP 5A. Inspect the ICON™ cab harness 14 pin connector pins.	No damaged pins
	STEP 5B. Check the cab harness and switches for a short circuit to ground.	More than 100k ohms
	STEP 5C. Check the ICON™ lamp for a short circuit to ground.	More than 100k ohms
	STEP 5D. Check the ICON™ parking brake switch for a short circuit to ground.	More than 100k ohms

STEPS		SPECIFICATIONS
STEP 6.	Check the ICON™ interlock components.	
	STEP 6A. Check the ICON™ neutral position switch.	More than 100k ohms
STEP 7.	Clear the fault code.	
	STEP 7A. Disable the fault code.	Fault Code 767 cleared

STEP 1. Read the fault codes.

STEP 1A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect the ICON™ electronic service tool.

Action	Specification/Repair	Next Step
	Fault Code 767 active.	2A
	Refer to Inactive or Intermittent Fault Codes, Procedure 019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html).	7A

STEP 2. Check the ICON™ idle control module.

STEP 2A. Check the ICON™ idle control module A connector for damaged pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module connector A from the ICON™ idle control module.

Action	Specification/Repair	Next Step
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• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	2B
	Repair the damaged pins • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html). • Repair or replace the OEM wiring harness as necessary.	7A

STEP 2B. Check the hood tilt switch for a short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ hood tilt switch.

Action	Specification/Repair	Next Step
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• Measure the resistance from pin B of the ICON™ tilt switch to engine block ground. • Measure the resistance from pin A of the ICON™ tilt switch to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	2C
	Replace the hood tilt switch. Refer to Procedure 019-298 (/qs3/pubsys2/xml/en/procedures/97/97-019-298.html).	7A

STEP 2C. Check the ICON™ components for a short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module A and B connectors from the ICON™ idle control module.
- Set the vehicle parking brake.
- Put the vehicle in neutral.
- Close the hood tilt switch or bypass the switch.

Action	Specification/Repair	Next Step
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Measure the resistance from pin 5 in the ICON™ idle control module A harness connector to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms Replace the ICON™ idle control module. Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html).	Repair Complete
		3A

STEP 3. Identify the ICON™ system.

STEP 3A. Identify whether the ICON™ system is an Aftermarket or an OEM.

Conditions : None		
Action	Specification/Repair	Next Step
Refer to Procedure 209-017 (/qs3/pubsys2/xml/en/procedures/97/97-209-017.html).	ICON™ system is Aftermarket	4A
	Check the OEM wiring harness for a short circuit to ground in the lamp output circuit.	7A

STEP 4. Check the ICON™ engine harness.

STEP 4A. Inspect the ICON™ engine harness 14-pin connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the 14-pin pass-through connector at the vehicle bulkhead.

Action	Specification/Repair	Next Step
• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	4B
	Repair the damaged pins • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/97/97-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html). • Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).	7A

STEP 4B. Check the ICON™ engine harness for a short to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module A connector from the ICON™ idle control module.
- Disconnect the 14-pin pass-through connector at the vehicle bulkhead.

Action	Specification/Repair	Next Step
Measure the resistance from pin 5 in the ICON™ idle control module A harness connector to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	5A
	Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).	7A

STEP 5. Check the ICON™ cab harness.

STEP 5A. Inspect the ICON™ cab harness 14 pin connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the 14-pin pass-through connector at the vehicle bulkhead.

Action	Specification/Repair	Next Step

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	5B
	Repair the damaged pins • Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the cab harness. Refer to Procedure 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html) • Replace the cab harness. Refer to Procedure 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).	7A

STEP 5B. Check the cab harness and switches for a short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the 14 pin cab harness connector.
- Set the vehicle parking brake.

Action	Specification/Repair	Next Step
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• Measure the resistance from pin J of the 14 pin cab harness connector, cab harness side, to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	6A
		5C

STEP 5C. Check the ICON™ lamp for a short circuit to ground.

- Conditions :**
 - Turn keyswitch OFF.
 - Disconnect the ICON™ lamp connector from the cab harness.

Action	Specification/Repair	Next Step
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• Measure the resistance from pin B of the ICON™ lamp connector to engine block ground. • Measure the resistance from pin A of the ICON™ lamp connector to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	5D
	Repair or replace the ICON™ lamp • Repair the ICON™ lamp. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html) or 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html). • Replace the ICON™ lamp. Refer to Procedure 019-046 (/qs3/pubsys2/xml/en/procedures/99/99-019-046.html).	7A

STEP 5D. Check the ICON™ parking brake switch for a short circuit to ground.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ parking brake switch.

Action	Specification/Repair	Next Step

• Measure the resistance from pin A of the parking brake switch connector to engine block ground. • Measure the resistance from pin B of the parking brake switch connector to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms Replace the cab harness. Refer to Procedure 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).	7A
	Replace the parking brake switch. Refer to Procedure 019-299 (/qs3/pubsys2/xml/en/procedures/97/97-019-299.html).	7A

STEP 6. Check the ICON™ interlock components.

STEP 6A. Check the ICON™ neutral position switch.

Conditions :

- Turn keyswitch OFF.
- Disconnect the neutral position switch.

Action	Specification/Repair	Next Step

• Measure the resistance from pin A of the neutral position switch to engine block ground. • Measure the resistance from pin B of the neutral position switch to engine block ground. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).	7A
	Replace the neutral position switch. Refer to Procedure 019-297 (/qs3/pubsys2/xml/en/procedures/97/97-019-297.html).	7A

STEP 7. Clear the fault code.

STEP 7A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Cycle the keyswitch to verify the fault code is inactive.	Fault Code 767 cleared	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting charts

